Introduction to Scientific Research

(Biostatistics and Scientific Research)

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Second semester- Third year

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Week-1

Outlines

- Search and Search Engine
- Search vs Research
- Scientific Research
- Scientific Publishers
- Scientific written communication
- General Tips for scientific Writing
- Elements of Journal Article

What is “Search”

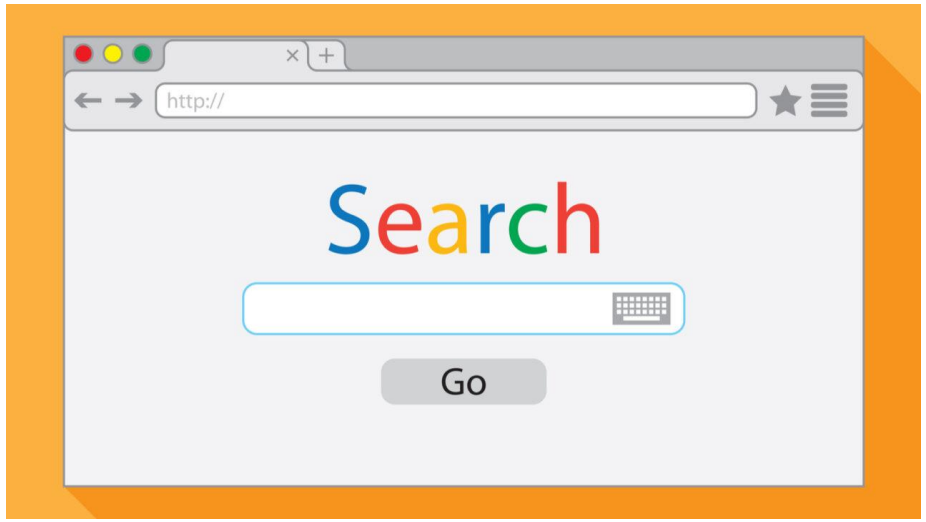
Search is:

- An act of **finding something** which has been lost before or missing
- A function or process of finding letters, words, files, web pages, or other data.
- Many operating systems, programs, and websites contain a search or find feature to locate data.

What is the difference between search and find?

- **Search:** you want to look for text in multiple documents or pages.
- **Find:** you try to find text in the current document or web page.

What is “Search Engine”



Definition:

- It is a **software program** that helps people find the information they are looking for online using keywords or phrases.

The search engine could be through:

- Web page search engine
- Image Finder
- Video Finder
- File Finder

Alternative Search Engines to Google

- ChatGPT
- Search.co
- Bing
- DuckDuckGo
- Yep.com
- You.com
- Boardreader
- BrandMentions
- CC Search
- WolframAlpha
- MetaGer
- Gigablast
- Quantcast
- Crunchbase
- Yandex
- Swisscows
- StartPage
- OneSearch
- Ekoru
- Twitter
- YouTube
- Oscobo
- Infinity Search
- Don’t Get Tunnel Vision
- Yahoo!
- Ask
- Baidu
- Brave
- WolframAlpha
- Ecosia
- Qwant
- Search Encrypt
- SearX
- Gibiru
- Disconnect Search
- Lukol

Academic Database Vs Google Scholar

Academic Database	Google Scholar
Database	**Search Engine**
Information is organized and standardized	Filter Google to pick out scholarly information
Searches for scholarly articles and other formats according to the indexed field (e.g., **keyword, date, limit by language, peer review, format, etc**.)	Searches for scholarly articles, theses, books, abstracts, according to " **Keyword**"
More limiting functions (e.g., research area, country, document type, etc.)	**Few limiting** functions (e.g., sort by relevance, date)
Niche type databases available (well organized)	Searches in academic publishers, professional societies, online repositories, universities and the free web
Paid by University on subscription basis for students	It does not have everything, and **not everything is free**!
PubMed Scopus Clarivate Web of Science ScienceDirect DOAJ JSTOR EBSCO	Google الباحث العلمي

Search vs Research

Parameter	Search	Research
Meaning	Search is the process of **looking for Something** in a **non-systemic** manner	Research is a process of reaching a **conclusion** in a **systemic** manner
Aim	To **find** something	To **reveal** new information and establish a correct conclusion based on that
Method	Search does **not** include **structured** techniques	Research includes the collection of **information, verification, analysis, and deduction**
Time Span	**Short**	**Long and complex**
Qualification	**NO**	**Yes**
Control	**Not monitored** or regulated	**Monitored** or regulated by someone other than the researcher
Dependency	**Not dependent on research**	**Research is dependent on search**

Scientific Research Road

What is Scientific Research?

Claim + Evidence + Reasoning ⟹ Explanation

Phenomenon
or
Problem

Questions?

Observations

Experiment

Probable
or
Absolute

Attention!!!

«مع أن الهدف من العلم هو حل المشاكل، إلا أنه يخلق المزيد منها»

ابراهام فلكسنر

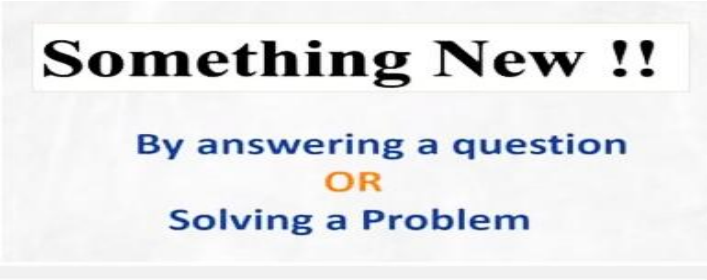
Scientific Research

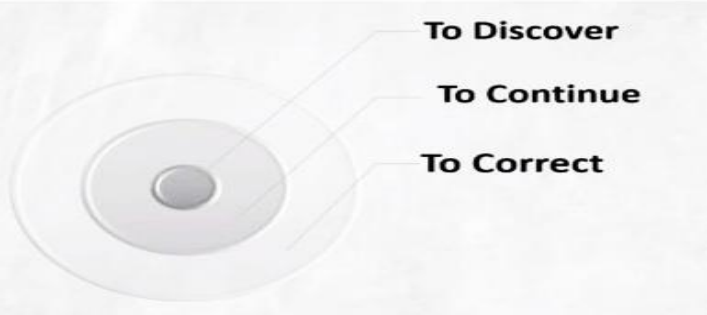
✓ Scientific research is investigating and acquiring or expanding our understanding

Philosophy of scientific research

- Filling knowledge gap
- Solve problem

Mix between two philosophy





Importance of Scientific Research

For World

- ✓ Helping us understand how things work
- ✓ Finding solutions for world problems
- ✓ Enhancing quality of life
- ✓ Devastating the world

For Us

- ✓ To Get a degree
- ✓ To improve skills
- ✓ To discover .....
- ✓ To make a useful contribution

Key features of Scientific Research

1. **Systematic:** Research involves a systematic and structured approach to investigation and inquiry
2. **Objective:** Research aims to be objective, unbiased, and free from personal opinions
3. **Empirical:** Research relies on empirical evidence, which is evidence based on observation or experience
4. **Theoretical:** Research is often guided by theoretical frameworks or models
5. **Analytical:** Research involves the analysis and interpretation of data, information, and evidence

Scientific Research Steps

Step	Requirements
1. Getting a research idea	- Critical thinking - Scientific background - Being a keen observer and reader - Identify a research problem or question
2. Reviewing the literature	- Ability to search Medical Electronic databases (PubMed, Cochrane, Google Scholar, EBSCO, EMBASE, SCOPUS, Web of knowledge, etc.)
3. Planning for the study (Research Design)	- Ability to write a study protocol - Selection of participants, data collection methods, and data analysis techniques.
4. Piloting the study	pilot sample
5. Gaining Ethical approval	Ability to comply with guidelines of your institution Ethics Committee or Institutional Review Board (IRB).
6. Study implementation	Clinical skills if interventional study
7. Data collection	Data collection tools e.g. Redcap
8. Data entry	Ability to use MS Excel, Google sheets, etc.
9. Data analysis	- Ability to interpret descriptive and inferential statistics. - Ability to run statistical tests on statistics software e.g. SPSS, R, STATA, SAS,… etc.
10. Writing manuscript	- Following standard reporting Checklists - Follow scientific writing rules - English language proficiency
11. Publication	- Presentation skills - Experience in Journal selection and answering reviewer's comments.

THANK
YOU